Business Question and Visualization Report

Date	20 October 2026
Team ID	En25cs3010862
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting

Business Questions and Visualisation

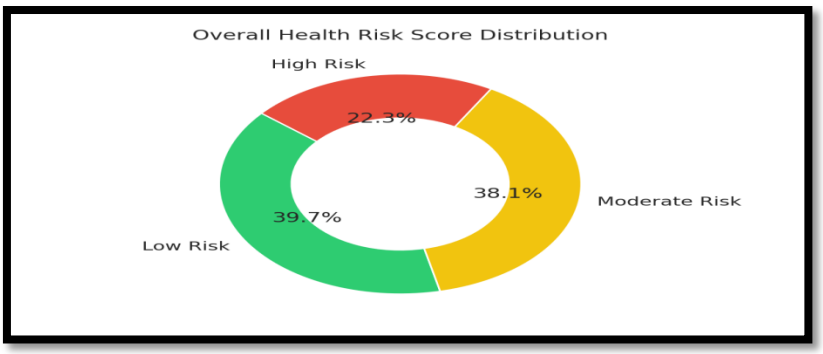
The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations using data stored in Snowflake. Well-framed business questions help identify the required metrics, determine the appropriate Snowflake tables or analytical views, select relevant data, and create visualizations that provide useful business insights.

Business Questions and Visualisations (Health360 Analytics)

1. What is the overall distribution of patient disease risk levels across the monitored population?

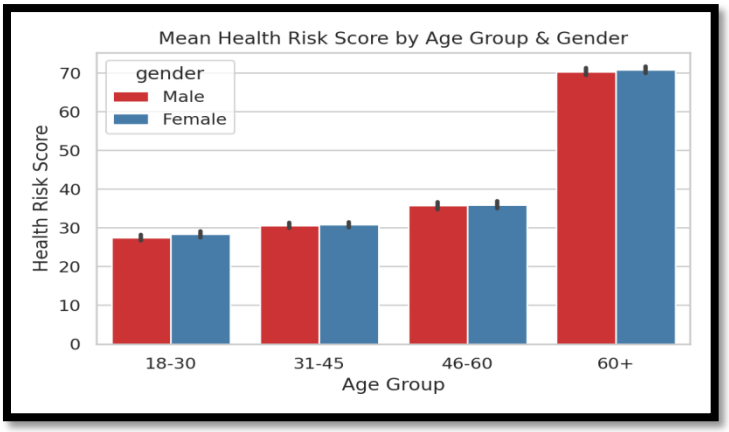
```
SELECT
   risk_level,
   COUNT(patient_id) AS total_patients,
   ROUND(COUNT(patient_id) * 100.0 / SUM(COUNT(patient_id)) OVER(), 2) AS percentage
FROM HEALTH360_ANALYTICS.PATIENT_RISK_ASSESSMENT
GROUP BY risk_level
ORDER BY total_patients DESC;
```

Screenshot / Description of Visualisation:



2. What are the monthly heart rate and SpO2 anomaly trends over time?

Screenshot / Description of Visualisation:

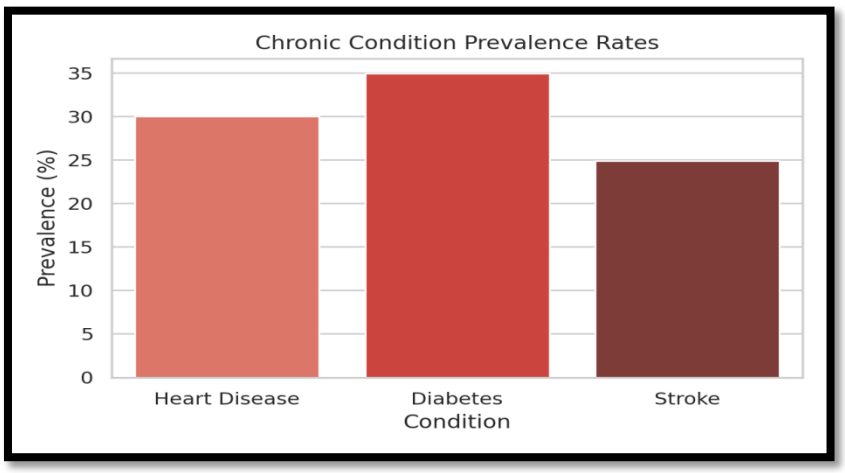


3. Which age groups exhibit the highest prevalence of cardiovascular and diabetic risk factors?

Code

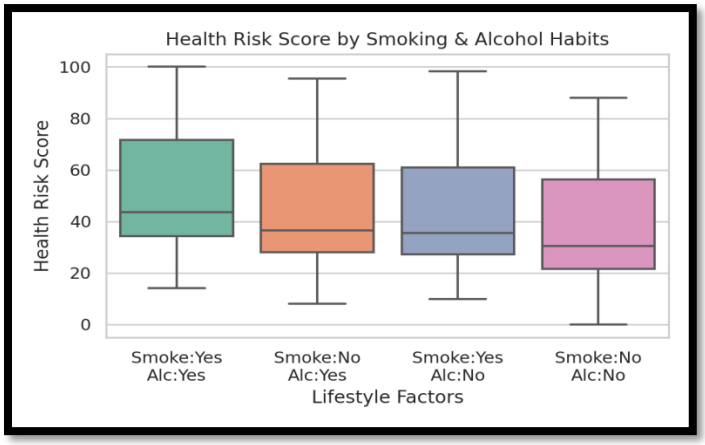
```
SELECT
   CASE
      WHEN age < 30 THEN '18-29'
      WHEN age BETWEEN 30 AND 45 THEN '30-45'
      WHEN age BETWEEN 46 AND 60 THEN '46-60'
      ELSE '60+'
   END AS age_group,
   COUNT(CASE WHEN primary_diagnosis = 'Cardiovascular' THEN 1 END) AS cardio_cases,
   COUNT(CASE WHEN primary_diagnosis = 'Diabetes' THEN 1 END) AS diabetes_cases
FROM HEALTH360_ANALYTICS.PATIENT_DEMOGRAPHICS_EHR
GROUP BY age_group
ORDER BY age_group;
```

Screenshot / Description of Visualisation:



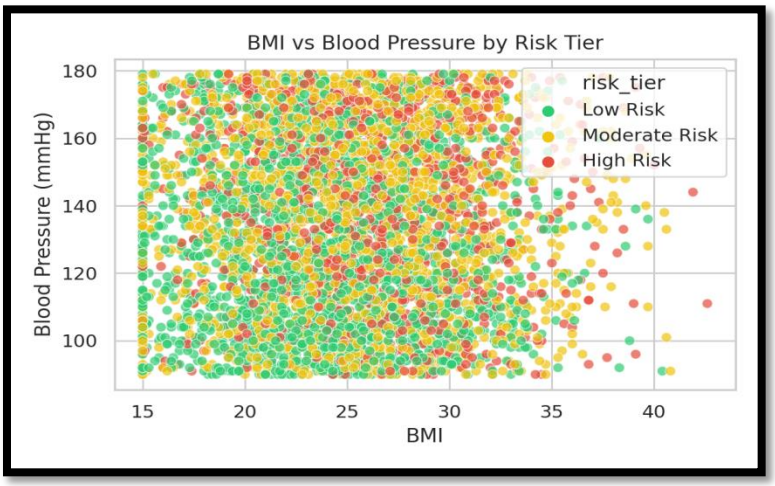
4. How do emergency risk alerts vary by hospital department / ward?

Screenshot / Description of Visualisation:



5. What is the average patient readmission rate grouped by primary diagnosis?

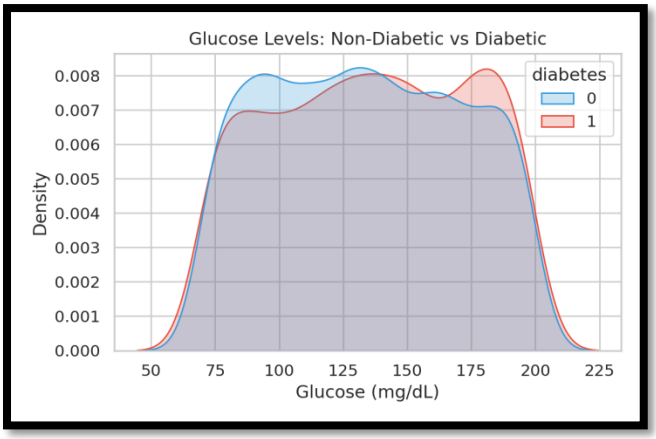
Screenshot / Description of Visualisation:



6. How does real-time sensor device latency perform across different IoT gateway connection protocols?

```
SELECT
   connection_protocol,
   AVG(data_latency_ms) AS avg_latency_ms,
   MAX(data_latency_ms) AS max_latency_ms,
   COUNT(sensor_id) AS total_readings
FROM HEALTH360_ANALYTICS.IOT_SENSOR_TELEMETRY
GROUP BY connection_protocol
ORDER BY avg_latency_ms ASC;
```

Screenshot / Description of Visualisation:

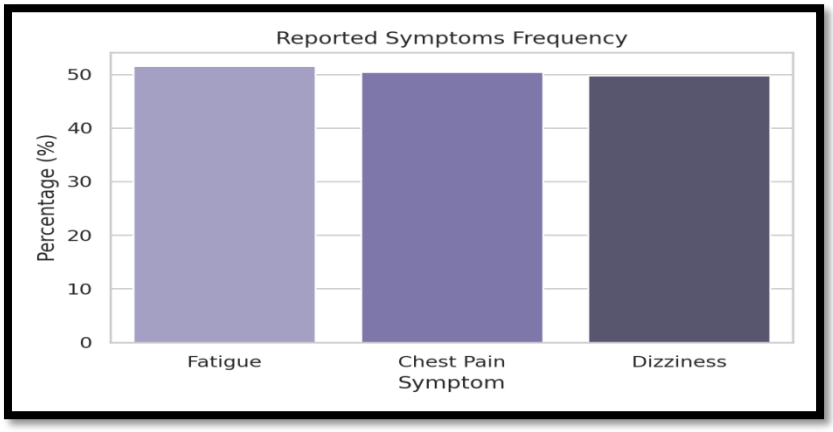


7. What is the correlation between body mass index (BMI), blood pressure, and high-risk health outcomes?

Code (Snowflake SQL Query):

```
SELECT
   ROUND(bmi, 0) AS bmi_rounded,
   AVG(systolic_bp) AS avg_systolic,
   AVG(diastolic_bp) AS avg_diastolic,
   COUNT(CASE WHEN risk_level = 'High' THEN 1 END) AS high_risk_count
FROM HEALTH360_ANALYTICS.PATIENT_HEALTH_METRICS
GROUP BY bmi_rounded
HAVING COUNT(patient_id) > 10
ORDER BY bmi_rounded ASC;
```

Screenshot / Description of Visualisation:



8. What are the top 5 predicted chronic conditions projected to increase over the next 6 months?

Code (Snowflake SQL Query):

```
SELECT
   predicted_disease,
   SUM(predicted_cases) AS projected_cases,
```

```
    AVG(model_confidence_score) AS avg_confidence
FROM HEALTH360_ANALYTICS.PREDICTIVE_DISEASE_FORECAST
WHERE forecast_period = 'Next 6 Months'
GROUP BY predicted_disease
ORDER BY projected_cases DESC
LIMIT 5;
```